

Jev and Typed Decision Models: An Empirical Survey of Calibration, Selective Control, and Open Implementations

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Abstract

In September 2026 TypeSafe AI released Jev, a hosted “System One” model that answers typed questions about a text state with a probability distribution over declared options instead of generated text. Within three weeks a literature formed around it: evaluations of the hosted model, open models and readouts that reproduce its interface, and systems that delegate bounded decisions to it. We review this evidence as it stood on 7 Oct 2026: 101 core studies, 1 peripheral study and 73 background references, read in full and extracted into 215 claim-level evidence records with locators, measurement scopes, sample sizes and model versions, together with an audit of 130 open models, benchmarks, applications and tools. Ten findings emerge. A type-valid answer can still be wrong: labels outweigh definitions, middle options absorb errors, and with an explicit “None” option hosted Jev rejected only 7% of arithmetic menus that lacked the right value. A native probability makes calibration measurable but not guaranteed, and a calibrated probability is not a coherent belief: probabilities of a statement and its negation do not sum to one, and interfaces disagree. Typed decisions read what the input states but do not reliably compute what it implies, and they are not secure against text placed in the state. The most consistent use is a bounded first pass with confidence-gated escalation; the strongest designs fix thresholds in advance and test on new workloads, yet audited savings can shrink sharply. Speed and cost figures come from incompatible scopes, system gains need attribution, open implementations are compatible but not equivalent, and the evidence base is young, clustered and unreplicated. We contribute a versioned evidence base, a synthesis that keeps incomparable numbers apart, an implementation and ecosystem audit, and a proposed evaluation protocol.

1 Introduction

Many software decisions are small and bounded: route a ticket, accept or escalate a judgment, pick the next tool, code a variable from a narrative, admit a network policy. Large language models can make such decisions, but they return strings that must be parsed, validated and retried, and the confidence they state is usually a verbalised estimate rather than a probability a program can threshold [1, 2]. Typed decision models change the contract. The caller declares the admissible answers; the model returns one of them with a probability for every option; no free text is generated, so a parse failure is impossible by construction.

TypeSafe AI’s Jev, released in early access on 15 September 2026, made this contract prominent [3]. Its launch materials describe a new class of models trained with “Reinforcement Learning for Calibrated Decisions” (RLCD) and report end-to-end responses of 70–500 ms at \$0.042 per million input tokens [3, 4]. Three weeks later the literature already spanned audits of option semantics and calibration, open models trained to copy the interface, and systems in judging, annotation, moderation, security, networks, databases, search, robotics, medicine and materials discovery. The claims range from schema validity to probability coherence, cost, latency and end-task success; the protocols range from 20 curated scientific choices to half a million police crash narratives.

Others have begun to map this literature. An early evidence audit reviewed the 28 papers posted between 19 and 24 September and derived a 14-item evaluation checklist [5]; a systematisation of knowledge examined semantic decision engines in network control loops [6]; a public survey draft places typed decisions in the history of classification, calibration and routing [7]; and community lists catalogue models, projects and benchmarks [8, 9]. This survey is an **empirical survey and evidence audit**: it reads every core study in full, re-reads studies when they post new versions, records each checkable claim with its locator and scope, and keeps incomparable numbers apart.

We keep three objects distinct throughout: (A) the **commercial model**, TypeSafe’s hosted Jev, whose architecture and training data are undisclosed; (B) the **paradigm**, Jev-like typed decision models and readouts built by others; and (C) **systems** whose results depend on typed decisions. A result about one is not evidence about another.

We ask six questions. **RQ1 (definition and lineage)**: how do runtime-defined finite-option decisions relate to classifiers, zero-shot labelling, rerankers, reward models and constrained generation? **RQ2 (prob-**

ability quality): what do returned probabilities and confidence mean, where do they hold, and are they coherent across related questions? **RQ3 (efficiency attribution)**: how much of a speed or cost gain comes from the readout, and how much from model size, batching, prefix sharing, caching, pruning or the network? **RQ4 (selective control)**: which decisions can be delegated, and when should software escalate, abstain or let code compute? **RQ5 (robustness and security)**: how do option names, order, definitions, missing options, language, injected text and poisoned checkpoints change decisions? **RQ6 (openness)**: what do alternatives release, and how much of the literature survives a common protocol?

Our contributions are concrete artefacts rather than claims of priority:

1. A versioned evidence base of 175 references and 215 evidence records. Each record carries a source locator, evidence type, test level, measurement scope, sample size and model version, with explicit reasons where a value is missing (§3).
2. A synthesis into ten findings, each listing supporting *and* qualifying evidence (§7). Study families are synthesised once, and nothing is pooled across incompatible scopes.
3. An implementation lineage of six readout families and an audit of the open ecosystem that separates code, weights, training code, evaluation material, data and raw predictions (§6, §9).
4. A catalogue of failure modes with sources and mitigations (§8), and a proposed evaluation protocol whose first experiments target binding, rejection, coherence and escalation together (§10).
5. A public website, generated README, BibTeX and CSV exports, all produced from the same data by scripts with validation and link checks.

2 Scope, definitions and what is documented

2.1 The interface

A caller sends a state (a string, JSON object or array of text values) and a map of typed questions; the model evaluates every question independently against the shared state and returns one answer per question [10, 11]. Three primitives exist. A **Choice** selects among named options, each defined by a description; the answer includes the chosen option, a probability for every option and a confidence value. The launch post states support for up to 255 options, with a two-stage procedure for high cardinality [3]. A **Score** places the answer on two to ten ordered, described levels; it returns a position that may fall between levels, probabilities per level and confidence, and the vendor warns against interpolating exact magnitudes from it [12]. A **Noul** returns the probability that a yes/no proposition holds and carries no separate confidence [11, 13].

At the cutoff the documented model was still `jev-1.13.0`; the aliases `jev-latest` and `jev-preview` both pointed to it, and the vendor advises pinning the versioned ID when thresholds are tuned to a version [4]. Input is text only; images, audio and video must first be converted to text or structured fields, and English is the primary training language [4, 10]. Visual and audio Jev-like models in the open ecosystem are independent projects, not TypeSafe releases.

2.2 Probability semantics

Probabilities and confidence are different objects. The documentation defines confidence as a statistic computed from the returned distribution and recommends thresholds tested on the user’s own data [13]; the exact statistic is not published. Empirically, one judging study found that native confidence detected errors about as well as the largest label probability (AUROC 0.745–0.876 across three benchmarks) [14], and a biosecurity audit read it as a chance-normalised maximum probability [15]. The vendor states that calibration is measured across groups of predictions and does not guarantee that an individual answer is correct [16]. Two independent sources observe that probabilities are returned on a 0.01 grid, which bounds calibration error for rare events from below and can put exactly zero on a correct answer [17, 18].

2.3 What is documented as weak, and what is not disclosed

The vendor maintains a list of known weak spots of `jev-1.13`. On its 2 October revision the list names literal reading, math and numbers (including counting and numeric representations), date and time comparison, indirection, large states full of irrelevant detail, adversarial content, contradictory instructions and criteria, Choice option order — the model “leans toward the option that comes first” — and generation [12]. Several studies below measure exactly these edges. The architecture, training data and objective are not published;

the documentation says one set of weights serves every account [4]. Community reconstructions exist and remain speculation about the commercial model [19].

2.4 Vendor claims as hypotheses

The launch headline “193.6× faster, 444.6× cheaper” comes from the vendor’s workflow evaluations, whose reference answers average two external frontier models and whose workflows were written by its capabilities team [3]. The vendor’s public evaluation dashboard now averages four structured workflows against reference labels from two frontier models: in workflow mode it lists Jev at 67.8% agreement for \$0.0004 and 0.4 s per case, against 74.1% for the best listed configuration and 73.1% for Opus 5, and every listed model agrees more often when the task is decomposed into a workflow than when it is asked as one prompt [20]. The plotted 0% type-error rate is analytical rather than empirical [3]. We record these figures as vendor claims, not as evaluations.

2.5 Names that collide

Several names mean different things. TypeSafe’s *System One* borrows Kahneman’s System 1 metaphor, which AI research had already adopted [21, 22]. *RLCD* in the Jev context expands to Reinforcement Learning for Calibrated Decisions and is unrelated to Reinforcement Learning from Contrastive Distillation (2023), a distinction one core study also draws [23]; open reimplementations of the Jev meaning now exist [24, 25]. *OpenJev*, *Open-Jev* and *open-jev* name at least eight unrelated projects: frozen-decoder readouts, among them SemIf under its former name, two DiffusionGemma readouts, a family of fine-tuned decision models, an in-browser runtime, and the open model of one paper’s title, JevLite [26–33]. Adjacency in name implies neither the same algorithm nor copying.

3 Review method

3.1 Search

The snapshot of 23 September 2026 ran 17 accepted arXiv API queries. They returned 225 hits and 197 unique records after de-duplication by unversioned arXiv ID (Appendix A). Queries combined the brand and interface vocabulary (jev, typesafe, “System One”, “typed decision”, “calibrated decisions”) with open-model names, survey terms and a date-bounded “decision model” query; arXiv `all`: searches metadata rather than full text, and that last query found a core study the brand-name queries missed [23]. A same-day increment added 19 expansion queries; queries whose phrases the API parsed so loosely that they returned thousands of records were rejected by rule.

Later increments re-ran every accepted and expansion query after the daily arXiv announcement and screened only records not seen before. The increment of 24 September added four core studies. The increment of 6 October retrieved 710 hits (444 unique records), of which 112 had not been screened: 74 became core studies, six were added as background or related reviews, and 32 were excluded with a recorded reason. Each run also refreshed the metadata of every included record; seven included studies had posted revised versions, which were re-read and their records updated. The increment of 7 October, run once that day’s listing had reached the API, retrieved 743 hits (458 unique records); ten of its 14 new records became core studies and four were excluded, and of three revised studies one had been substantially rewritten and was re-read. Every screening decision and reason is published with the data.

Repositories came from GitHub searches and from curated lists. For the lists we checked each entry against its own README or paper before inclusion: entries whose README did not mention Jev or a typed decision model were excluded, and general infrastructure or prior work listed for context was recorded as context only [8].

3.2 Eligibility

A study is **core** if it evaluates TypeSafe Jev, a clearly identified Jev-like typed-decision implementation, or a system whose contribution materially depends on such decisions. **Peripheral** studies fall within reach of the scope but rest on contested, self-reported claims; they stay visible but are not pooled. **Background** references are adjacent methods, related reviews and earlier work the interface descends from, each with a stated role. We excluded name matches (Japanese encephalitis, JEPA, unrelated uses of “TypeSafe” or “System One”, other expansions of RLCD) and work without a typed-decision link.

3.3 Reading depth and extraction

We read the core and peripheral studies in full, checking key tables and limitation sections; background references were checked against metadata and abstracts. Code, weight and data links reported by the studies were checked to resolve, and releases that a paper announces but that could not be found are recorded as “claimed, not located”. Repository evidence comes from READMEs and metadata at pinned commits, with selected source files for a few. Nothing was executed and no experiment was re-run.

Each checkable statement became an evidence record. A record stores its subject, text, headline and source locator, and an evidence type: author-reported experiment, vendor documentation, vendor-reported result or community report. It also stores the metric, value and baseline; the sample size, model version and hardware, each with a reason when unknown; the **test level** (model test, system test, hybrid, simulation); and the **measurement scope** (single request, amortized per question, batch, end to end, simulation, author estimate, vendor claim). Each record links to the findings of §7 as *supporting* or *qualifying* evidence. Openness is recorded as six independent fields per study: code visible, weights available, data available, raw predictions available, recomputable, independently reproduced.

3.4 Synthesis and its limits

Tasks, metrics, binning and scopes are heterogeneous, so we pool nothing and rank nothing. Findings are stated at the strength their records allow, with qualifying records listed beside supporting ones. Studies that share authors are flagged as study families and read together: two edge and RAN studies by one group (which also wrote the network SoK), three studies by Rafe and Das, an agent architecture and an attack study by Wu and Lim, two studies by Yang and Zhao, a Turkish benchmark whose author also built a model it ranks, and a reproduction of Laya and a judge evaluation of CLM by one author [17, 34–45]. The review was carried out by one AI-assisted reviewer and is not a registered or PRISMA-compliant systematic review; §12 lists what this implies.

4 Related reviews and positioning

Typed Decision Models: An Early Evidence Audit reviews the 28 studies posted between 19 and 24 September 2026 [5]. It concludes that the typed readout had not yet shown an accuracy advantage over comparable label-probability readouts and that the clearest gains were latency and cost, and it turns recurring weaknesses into a 14-item evaluation checklist. Our reading of the same window agrees; this survey extends it by two further weeks, records evidence claim by claim, re-reads studies when they are revised and audits the open ecosystem.

SoK: Semantic Decision Engines in Network Control Loops systematises 139 paper families by decision interface, execution path and check ownership [6]. Of 50 families that claim their engine fits a control loop or time budget, four support the claim with matched measurement, and bounded tests show that queueing can turn a deadline met by every isolated request into one met by none. It is written by the authors of two edge and RAN studies reviewed here, so we treat its examples of those studies as part of that family.

Decisions, Not Tokens, a public working draft dated 21 September 2026 and unchanged since, treats machine-native decision models broadly, from classical classifiers to Jev, with a five-dimensional taxonomy and four evaluation layers; its bibliography at the pinned commit cites none of the core studies reviewed here [7]. Community catalogues list models, projects, benchmarks and papers with one-line descriptions [8, 9, 46]; they were indispensable for discovery, and every entry we use was traced to its own source. A perspective essay proposes a conditional Jevons hypothesis: much cheaper machine evaluation may raise its use where demand is latent, while shared criteria and shared errors can scale mistakes [47].

Adjacent surveys cover uncertainty quantification and calibration in LLMs [2], the move from System 1 to System 2 reasoning [22], efficient reasoning under budgets [48] and decision-focused learning [49]. None addresses typed decision APIs.

5 Lineage and an analytical frame

5.1 Lineages

Typed decisions recombine long-standing ideas. **Discriminative classification** with bidirectional encoders [50, 51], few-shot classifiers [52] and generalist lightweight classifiers [53, 54] are the natural baselines; zero-shot labelling via label descriptions anticipated runtime-defined options [55]; cross-encoder rerankers score a supplied candidate without generation [56]; and controlled comparisons of encoder, autoregressive and

Table 1: Five places a claim can live: an analytical organisation, not a model architecture.

Stage	Question	Evidence to look for	Studies
Decision contract	What exactly is being asked, and what may the answer be?	Type validity; name and order invariance; closed-set failure when the right option is missing.	28
Inference & readout	How is a distribution over the declared options produced?	Same-backbone ablations; readout geometry; what is and is not disclosed.	32
Probability & calibration	What does a returned probability or confidence mean, and for which population?	Per-task reliability, high-confidence errors, recalibration on held-out labels.	46
Selective control	When should software act, ask, escalate or abstain?	Coverage at fixed risk; escalation rate; thresholds chosen on a separate selection set.	37
Downstream use	Does the whole workflow get better, cheaper or faster?	System-level baselines; module attribution; matched measurement scope.	60

diffusion classifiers map the trade-offs typed models inherit [57]. Reward models that score a whole response with one scalar [58] and safety classifiers that gate a model on a fixed policy [59, 60] are deployed non-generative judgment heads. **Calibration** and temperature scaling [61], the failure of calibration under shift [62], the loss of calibration after post-training [63], self-knowledge [1] and semantic uncertainty [64] frame every probability claim, and reinforcement learning with calibration rewards is the closest published relative of calibration-oriented decision training [65–67]. **Selective prediction** [68, 69], learning to defer [70] and conformal methods [71, 72] supply the theory of acting on confidence, and decision-aware calibration measures what matters downstream [73–76]. **Structured output** research shows that constrained decoding guarantees validity but not semantics [77–79], that format restrictions can cost reasoning accuracy [80], that declarative typed signatures can be compiled into prompts [81], and that schema compliance needs its own benchmarks [82, 83], while forms that demand an answer invite invented ones [84]. **Serving** work explains much of any speed-up independently of the readout: paged KV caches and prefix sharing [85, 86] and parallel diffusion decoding [87, 88]. **Judges and reward models** [89–91], **routing and cascades** [92–95] and **label and format sensitivity** [96–98] complete the background.

5.2 Five places a claim can live

We read each study by where its claim sits on the path from a declared question to a workflow outcome (Table 1). This is an analytical organisation for comparing evidence. It is not the internal architecture of any model, and we do not claim it as the first such taxonomy.

6 Implementation families

Implementations share the request and response shape, not the mechanism that produces the probabilities. We distinguish six families by their readout (Table 2).

Hosted service. TypeSafe’s Jev documents its interface, prices, limits, versions and known weak spots. It is observable only as a black box that can change behind aliases.

Encoder decision heads. Laya uses ModernBERT-large and mmBERT-base checkpoints with a router [99]; an independent reproduction matched its reported accuracy but found it uniformly under-confident, fixed by one temperature [44]. The family has grown in three directions. Encoder–decoders: Bongard reads the state once with a bidirectional T5Gemma encoder and lets decoder branches answer many questions, reaching 78.05% on DecisionBench and 32 questions about one state in 221 ms on one GPU [100]. Languages: open encoders now exist for Chinese, Japanese and Turkish; the Chinese one slightly exceeded hosted Jev on a general Chinese subset with a third of its calibration error, and the Turkish one scores each option blind to the others, which makes it order-invariant by construction [43, 101, 102]. Domains: biomedical sentence encoders converted to decision heads, where retrieval pre-training helps one head and hurts another [25], and, outside language, small heads on one pass of a frozen interatomic potential that answer typed questions about candidate crystals [103]. A contrastive variant scores state–action pairs with two small heads on a frozen 8B embedding model behind a Jev-compatible API [104]; zero-shot it judged near chance in a preregistered study, although its decisions ignored answer order [45]. The readout geometry matters: in open models the short option label outweighed its written definition, and a single change to how options are rendered made three checkpoints exactly invariant [105].

Frozen decoder readouts. An unmodified instruction-tuned model, prompted once, already works as a decision model. Read through bracketed numeric identifiers, a frozen Qwen3.5-4B resolved 81.4% of the

Table 2: Six readout families. Compatible interfaces do not imply equivalent mechanisms.

Family	Principle	Public	Boundary
Hosted System One service	A closed, hosted model returns a distribution over declared options for each typed question about a shared text state.	Interface, primitives, pricing, limits, versioning and known failure modes are documented; architecture, training data and the exact confidence statistic are not.	Black-box evaluation only; the served model can change behind aliases; text-only input.
Encoder decision heads	A bidirectional encoder, or an encoder-decoder, reads state, question and options; a trained head scores each option and a softmax is taken over the legal set.	Weights and inference code for most members; several release training code; the readout geometry (marker token, span mean, blind per-option scoring) changes name and order sensitivity.	Short context windows; option labels can dominate written definitions; adjacent to GLiClass, SetFit and NLI zero-shot classifiers.
Frozen decoder option readout	An unmodified causal LM is prompted once and the next-token logits of short option identifiers are renormalised over the declared options, optionally debiased over label priors and option rotations.	Fully open and training-free; a frozen 4B model read this way matches community fine-tunes on public JevBench items — the natural baseline for any trained decision model.	Raw logits are order- and label-sensitive; one prefill cannot reason; rotation averaging multiplies cost by the number of rotations.
Fine-tuned decoder decision models	A decoder is adapted (LoRA or full fine-tuning) so that a designated position or pointer head yields calibrated option probabilities in one pass.	Many release weights, training code and evaluation data; lineages exist (this-that-model is adapted from decider-2b; SecJev builds on Kev); task-specialised variants beat general ones on their own tasks.	Gains may come from task coverage in training; benchmarks are often built by the same teams; derived checkpoints are not independent methods.
Diffusion structured readout	A discrete diffusion LM is seeded with the answer template; after one denoising step, the distribution at each answer slot is read directly.	Open servers with a Jev-compatible wire API; depends on request extensions proposed in an unmerged vLLM PR.	A different mechanism from last-token decoder readout; custom entropy-based confidence.
Generative & constrained adapters	A general LLM is asked (often under JSON-schema constraints) to emit a decision and probabilities, which are validated and returned in Jev-shaped responses.	Wire compatibility for controlled comparisons; probabilities are generated or verbalised, not read from logits.	Interface compatibility is not mechanism equivalence; retries, parsing and self-reported probabilities remain.

public JevBench items, matching community Jev-style models built on the same backbone [106]. Its weakness is bias: reversing the option list changed the answer on about a third of 20-option items across eleven models, and averaging over cyclic rotations roughly halved the flips at the cost of one prefill per rotation [107]. Many open projects implement this family [26, 108–110], and behaviour-tree agents score complete candidates from one shared KV cache [111].

Fine-tuned decoder models. Kev, decider, this-that-model, NanoJev, Bespoke Nimble, JevLite, Open-Jev and others train a decoder so that a designated position or pointer head yields option probabilities in one pass [19, 31, 33, 112–115]; a Thai–English model replaces the language-model head with a slot softmax [116]. Task specialisation now beats generality on the specialists’ own tasks: a security-specialised 0.8B model outscored a general 9B by 20.5 points [117], a search-decision model beat hosted Jev on query rewriting while trailing on routing [118], and a browser-action model completed more than twice as many real-website tasks as Jev in the same agent although the two were close on single-step benchmarks [119]. Multimodal variants read screens, photos, pixels or medical images [120–123]. Distilling an open LLM’s reasoning into a single first-token decision produced a 27B model on par with Jev at about 0.1 s per query [124]. Architectural work makes candidates nearly order-invariant by blocking attention between them, cutting permutation flips for one model from 18.2% to 1.2% [125], while encoding candidates independently for caching cost rule sensitivity [126]. Open reimplementations of calibration-oriented training disagree: one beat SFT, STaR and GRPO at selective prediction [24]; another trailed plain cross-entropy by 2.5–3.0 points because of its reward normalisation [25]. Looping adds computation without parameters: a looped 1.4B decision model gained 13.5 points over a single-pass model of the same shape, data and recipe [127], whereas a project that loops only the top two layers of a single-pass model shows no accuracy gain over its backbones in its own table [128]. A 4B model with Laya’s decision head passed criteria registered before its test on randomised-trial questions [129].

Diffusion structured reads. djev and OpenJev seed a discrete diffusion model’s canvas with the answer template and read each slot after one denoising step [29, 130]; another harness reads typed JSON decisions from DiffusionGemma without training [30].

Generative adapters. TypeSafe’s official adapter routes typed questions to general LLM APIs for comparisons [131]; LocalJev asks a model for JSON probabilities and calls the result wire-compatible but “not

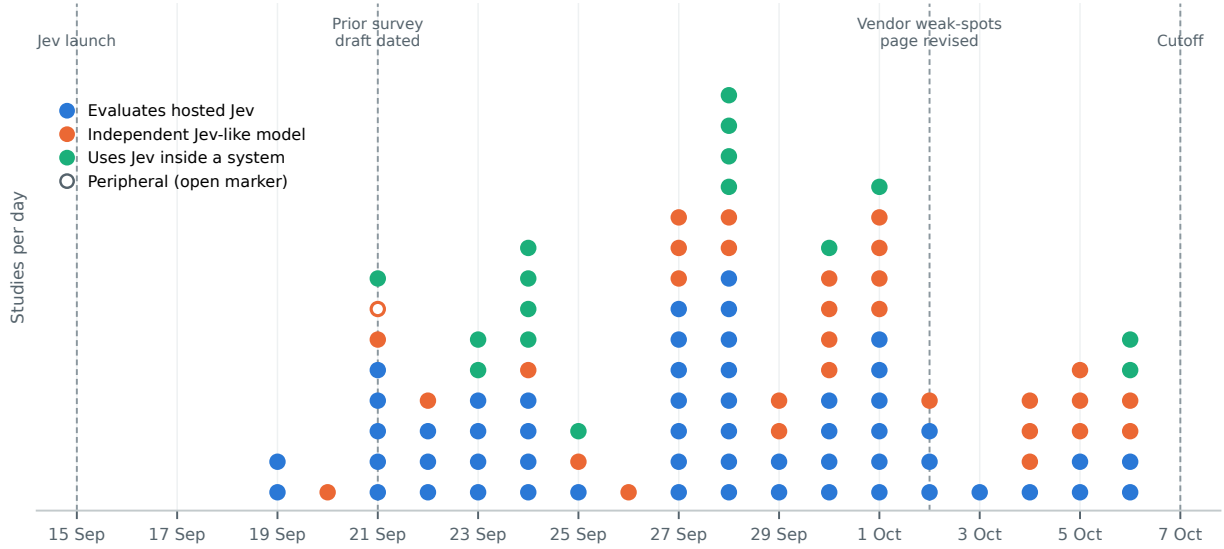


Figure 1: Submission dates (arXiv v1) of the core and peripheral studies, one dot per study stacked by day and coloured by primary relationship to Jev, against context events.

mathematically equivalent” to a logit read [132].

The implication for RQ1 and RQ6 is direct. The interface can be reproduced by many mechanisms whose probabilities mean different things, and the readout leaves a measurable trace: for prefill-only decision models, accuracy after a change of candidate menu can be predicted from the cached first pass, which is not true of same-scale generative models [133]; frozen readouts are the baseline any trained decision model has to beat; and none of these projects reveals how the commercial model works.

7 Evidence synthesis

7.1 The evidence base

The core studies appeared between 19 September and 6 October 2026, beginning four days after the launch (Figure 1). Relationships overlap: 61 studies evaluate hosted Jev, 52 build or evaluate independent Jev-like models, and 20 embed typed decisions in a larger system (Table 3). In all, 75 studies call hosted Jev, as the object of study, a baseline or a component; 63 state the version they called and 12 state none. Every core study is an arXiv preprint.

Table 3: Core and peripheral studies (* peripheral; † same study family). Headline values are author-reported.

Study	v1 (2026)	Test level	Version	Headline (as reported)	Ref.
Replacing LLMs with Jev at the Edge †	09-19	Hybrid	jev-1.13.0	Median decision latency −22.7% to −64.5% vs the fastest LLM	[34]
Intent at RIC Timescales †	09-19	Hybrid	jev-1.13.0	Meets the 1 s RIC budget on 99.8% of calls (LLMs: 17.9%, 0%)	[35]
this-that-model-1.0	09-20	Model test	1.0; not reported	30.9 ms per decision on one laptop GPU; 32 decisions/s	[113]
Open-Jev on CallScreen-Bench	09-21	Model test	paper v1 (three seeds)	JevLite ensemble AUROC .974, ECE .052; no false alarms	[33]
Jev-Mem	09-21	System test	not reported	LoCoMo judge 0.777 (+11.0%); build 158 s; 0.93 s/query	[134]
Calibrated Decisions at Scale †	09-21	System test	jev-1.13.0	499,500 narratives screened for \$25.23; 195,857 fully coded	[17]
JEVQA	09-21	Model test	jev-1.13	PLCC vs VMAF: 0.737 (metadata) → 0.824 (+bit-stream, pixel)	[135]
Decision Models for CSS Annotation	09-21	Model test	typesafe/jev1.13	Behind best LLM on 14/15 tasks (−11.6 F1) at 44× lower cost	[23]
Pregnancy in Crash Narratives †	09-21	Model test	jev-1.13.0	Population-scale reading audited by blind coders: sensitivity 0.999	[36]

Study	v1 (2026)	Test level	Version	Headline (as reported)	Ref.
Jev for Scientific Decisions	09-21	System test	Jev 1.13	Full semantic correctness; median 0.335 s, p95 0.442 s	[136]
Universal Fractal Decision Map *	09-21	Model test	not reported	Self-reported 81.65% on public JevBench items at 7.08 ms	[137]
Visual Jev	09-22	Model test	paper v1	Macro accuracy 0.706 $\rightarrow$ 0.761 after post-training (seen families)	[120]
REFLEX with Jev †	09-22	System test	jev-1.13.0	95% success with 1.12 strong calls/task vs 88% and 4.10 (−72.7%)	[38]
JEV-as-a-Judge	09-22	Model test	jev-1.13.0	Within 3 points of GPT-6 where the verdict can be read off the text	[14]
Type-Safe Is Not Error-Free	09-22	Model test	not reported	Hosted Jev AUROC .8146 $\rightarrow$.5806 after name–rubric swap; 0% type errors	[138]
JEV-Star	09-23	System test	Jev 1.13	4/4 full-game wins incl. two vs Lv7 with GPT-6 plans; Jev-only never expanded	[139]
KITE	09-23	System test	jev-1.13.0	1.7% flagship anchors cut effect error 41%; decision gain 0.27 $\rightarrow$ 0.39	[140]
Jev as a Radiology Report Judge	09-23	Model test	jev-1.13.0	Kendall τ 0.573 / 0.398 with expert error counts; beats open NLI	[141]
Jev on Contract Inference	09-23	Model test	jev-1.13.0	\$0.000228 and 1.24 s per contract — lowest of ten models	[142]
NumericJev	09-23	Model test	jev-1.13	Range refinement: 83.4% within 5% vs 80.5% direct choice	[143]
Decision Hijacking †	09-23	Model test	jev-1.13.0	Injection shifts probabilities (+0.043) but rarely selects the target (1.8%)	[39]
Harness Tokenomics	09-24	Simulation	jev-1.13.0	Emulated router saves 13–21% of coding-agent spend	[144]
Pentest Decision Layers	09-24	System test	jev-latest	One run each: 5 min faster, different severities — not a controlled test	[145]
PixelJev †	09-24	Model test	—	Typed readout and generation gain equally after adaptation	[40]
Just Ask Jev	09-24	Model test	jev-1.13.0	Zero-shot failure detection: median AUROC 0.886 at 1/63 of judge cost	[146]
Jev as a Rubric Judge	09-24	Model test	jev-1.13.0	Parity on most rubric criteria at 1/16–1/325 of the cost	[147]
Jev-Mobile	09-24	System test	not reported	AndroidWorld 79% (vs 84% step-wise) at 73% lower API cost	[148]
Jev in the Wild	09-24	Not applicable	not reported	Ecosystem: judgment 77%, scoring 52%, action selection 31% of projects	[149]
JevOut	09-24	Model test	jev-1.13.0	Natural context flips 61% of correct decisions (vs 2% neutral)	[150]
Biosecurity Audit	09-24	Model test	jev-1.13.0	Confidence = chance-normalised max probability; pooled ECE 0.034	[15]
LAVOIR	09-25	Model test	—	Learned “when to ask” policy $\approx$ oracle (AUC 0.80 vs 0.80)	[151]
JevSoup	09-25	System test	jev-1.13.0	Jev-routed LoRA composition: +0.2 to +1.2 points	[152]
JevAdvBench	09-25	Model test	jev-1.13.0	An appended opinion flips 12% and drains 38% of confident answers below the gate	[153]
PACT	09-26	Model test	—	Open recipe holds up; extra terms buy robustness, not accuracy	[154]
Probability Axioms	09-27	Model test	jev-1.13	P(X) + P(not X) misses 1 by 0.064 (open LLM readout: 0.293)	[155]
Agent Security Decisions	09-27	Model test	jev-1.13	Safe automation coverage: 7.6% (symmetric) $\rightarrow$ 47% (separate thresholds)	[156]
Speech Neuroprosthesis Rescoring	09-27	Model test	jev-latest	Rescoring: WER 7.5% vs 7.8% for 7B LLMs, no GPU needed	[157]
COGNIT-Guard	09-27	Model test	—	Adapted open guard: AUC 0.83 $\rightarrow$ 0.997, with an OOD “alignment tax”	[158]
Typed Decisions for 5G Control	09-27	System test	jev-1.13.0	Hosted Jev never meets 100 ms near-RT deadlines (warm p50 281 ms)	[159]
Laya Reproduction †	09-27	Model test	—	Open Laya checkpoint under-confident (gap −0.214); one temperature fixes it	[44]
Probability Contracts	09-27	Model test	jev-1.13.0	Two interfaces, same question: different decision on 32.8% of pairs	[160]
JET	09-27	Model test	Jev 1.13	Local likelihood readout 85.1% vs Jev 89.1% on MMLU	[161]
Do Decisions Add Up?	09-27	Model test	jev-1.13.0	Staged decisions disagree with direct ones; Jev loses 22.9 points on CLINC150	[162]

Study	v1 (2026)	Test level	Version	Headline (as reported)	Ref.
Jev in Medicine	09-27	Model test	jev-1.13	Medicine: parity on abstracts, −22 points on diagnostic cases	[163]
Video Anomaly Readouts	09-28	Model test	jev-1.13.0	Same evidence, different readout: +18 AP on one dataset, none on the other	[164]
Selection vs Extraction Memory	09-28	System test	jev-1.13.0	One Jev selection call $\approx$ LLM-extraction memory at 1/3,061 the write cost	[165]
SeLMRoute	09-28	System test	not reported	Typed-question features route LLMs: 72.1% vs 69.2% best single model	[166]
Agent Trace Security Judge	09-28	Model test	not reported	Trace security: F1 77.8 vs 74.1 for the best LLM judge	[167]
JevVibe	09-28	Model test	jev-1.13.0	CWE triage: Top-1 0.46 vs 0.50 (frontier) at 1/56 of the cost	[168]
NavJev	09-28	System test	not reported	Navigation: 27% SR at 0.65 s/step; components, not just Jev, carry the gain	[169]
Source-Position Coherence	09-28	Model test	jev-1.13.0	Jev’s argument scores move with the stated source	[170]
Decide, Don’t Generate (ABSA)	09-28	Model test	jev-1.13.0	SemEval ABSA: best regression error with 488 fitted coefficients	[171]
Sys1Cal-v1	09-28	Model test	not reported	Known-probability test: Noul 0.92, Score 0.89, Choice 0.76	[172]
Mnemon	09-28	System test	jev-1.13	Memory agent: LoCoMo 91.7% with Jev judging evidence (AUC 0.94)	[173]
Koa-action	09-28	Model test	—	Single-token LLM classifier: 85.5% at 0.53 s, flat across label spaces	[174]
Dyad	09-28	Model test	Jev 1.13	Native typed-decision head: 86.2% vs Jev 85.0% on JevBench	[175]
Training-Free HAR	09-28	Model test	jev-1.13.0	Sensor activity recognition from text: macro-F1 $\leq$ 0.12	[176]
Cultural Values Audit	09-28	Model test	jev-1.13.0	Persona shifts reproduce 87% (English) vs 62% (Arabic) of a human difference	[177]
Chinese-Jev	09-29	Model test	not reported	Language-specific open encoder $\approx$ hosted Jev in Chinese, better calibrated	[101]
Benchmarking Jev (37 datasets)	09-29	Model test	jev-1.13.0	Ahead of a 27B open LLM on 27/37 datasets; pooled Choice ECE 0.028	[178]
Certo (Cacheable Rules)	09-29	Model test	—	Cacheable scoring loses rule sensitivity (recall@1 1.00 $\rightarrow$ 0.24)	[126]
Decision Gates Benchmark †	09-29	Model test	jev-1.13.0	Threshold for 5% risk still admits 31% of out-of-scope requests	[37]
Ordinal-Scale Bias	09-30	Model test	typesafe/jev-1.13-20260917	ANLI: Neutral absorbs 51% of errors, at every position	[179]
OpenJev-RLCD	09-30	Model test	—	Open RLCD: best selective prediction among SFT, STaR, GRPO	[24]
Bongard	09-30	Model test	—	Open encoder–decoder: 78.05% on DecisionBench, 32 questions in 221 ms	[100]
Network Traffic Classification	09-30	Model test	typesafe/jev-1.13-20260917	Packet numbers in, 28% accuracy out — trees trained on 8k records reach 70%	[180]
OmniMed-Jev	09-30	Model test	—	Same backbone, decision-native output: calibration error down up to 10 $\times$	[123]
Rejection Bottleneck	09-30	Model test	jev-1.13	Present 99%, rejected when absent 7% — though Boolean checks get 99%	[181]
JevSpawn	09-30	System test	jev-1.13.0	Agent with spawned finite actions: Maze 0.96 vs 0.52	[182]
Recommendation Reranking	09-30	Model test	jev-latest	Strong reranking quality; latency between local models and pointwise LLMs	[183]
AnyJev	09-30	Model test	—	Order flips on a third of 20-option items; rotation averaging halves them	[107]
Beyond Answer Confidence	10-01	Model test	jev-1.13.0	Confidence $\neq$ knowing: +0.21–0.33 overconfidence past the knowledge boundary	[184]
Jev-IDS	10-01	Model test	jev-1.13.0	Intrusion pilot: F1 0.859, 4.8 $\times$ faster than an LLM	[185]
Fast Models, Slow Evidence	10-01	Model test	jev-1.13.0	Hosted beats open on 9 of 11 harness decisions; routing at chance for both	[186]
Candidate-Independent Attention	10-01	Model test	—	Isolate candidates: order flips 18.2% $\rightarrow$ 1.2%	[125]
Code Owns the Simulation	10-01	System test	jev-1.13.0	Evaluates what is described; does not simulate what is not	[187]
HakemBench †	10-01	Model test	jev-1.13	Turkish typed decisions: Jev fourth of 16 at a tenth of the leader’s time	[42]

Study	v1 (2026)	Test level	Version	Headline (as reported)	Ref.
JEVDB	10-01	System test	jev-1.13.0	Semantic SQL: pruning + typed decisions finish 540K-pair joins	[188]
HydroJEV	10-01	Model test	jev-1.13.0	Pre-registered gate: a third of SCADA reviews of-flooded, accuracy kept	[189]
LLM-as-Jev	10-01	Model test	—	A frozen 4B LLM already matches community Jev-style models	[106]
SBERT2S1	10-01	Model test	—	Open RLCD recipe trails plain cross-entropy by 2.5–3.0 points	[25]
Labels Override Definitions	10-01	Model test	—	The label wins over the definition — and the cause is one rendering line	[105]
SecJev	10-02	Model test	—	Security-specialised 0.8B beats general 9B by 20.5 points	[117]
HateDecide	10-02	Model test	jev-1.13.0	Definitions move 28% of answers — without reliably improving them	[190]
Candidate Coverage and Transfer	10-02	Model test	jev-1.13.0	Rejection thresholds do not travel: 69% false rejection after transfer	[191]
JEVal and InnerJev	10-02	Model test	jev-1.13.0	Right mode, wrong mass: 89% placed where the truth is 36%	[124]
Wireless Decision-Making	10-03	Model test	not reported	Wireless control: 3.5–8.5× faster than LLMs; quality depends on the task	[192]
Hidden Risks of Jev	10-04	Model test	not reported	Typed outputs, hijacked inputs: up to 100% injection success	[193]
CrystalJev	10-04	Model test	—	One calibrated pass against a relaxation: as good or better at a thirtieth of the cost	[103]
SearchJev	10-04	Model test	jev-1.13.0	Task-specialised open model beats Jev on rewriting, trails on routing	[118]
Fiorillo	10-04	Model test	—	Registered test passed: ECE 0.017, macro-F1 0.92 against 0.87 for a 31B model	[129]
GraphDecide †	10-05	Model test	jev-1.13.0	Reads every edge; counts degree right 68% of the time	[41]
Political Science Replications	10-05	Model test	typesafe/jev-1.13-20260917	Same accuracy, no cost advantage at batch prices	[194]
ufakzeka-karar †	10-05	Model test	—	Order invariance by construction; sequential heads still flip 2–3%	[43]
CLM-as-a-Judge †	10-05	Model test	—	Open contrastive decision model judges near chance, but ignores answer order	[45]
SharedKV-BT	10-05	System test	—	Node-local candidates: 113/120 correct decisions against 90/120	[111]
Readout Stability	10-06	Model test	not reported	Menu changes are predictable offline for decision models, not for generative LLMs	[133]
SanSi	10-06	Model test	jev-1.13.0	Looping a small decision model: +13.5 points at 7.7× compute	[127]
System One Moderation	10-06	Model test	jev-1.13	Precedents help Jev pick the right rule (+5 to +22 points); Laya barely moves	[195]
Plans Change Answers	10-06	Simulation	—	Calibrated confidence prices query plans; overconfidence hides output errors	[196]
S1-MAS	10-06	System test	—	Laya coordinates agents with 78–94% fewer GPT tokens; random choices trail by 1.8 points	[197]
BOTTLED	10-06	Model test	jev-1.13.0	Agent-built artifacts: 94–97% of Jev’s quality at 2–25% of its projected cost	[198]

7.2 F1 • A type-valid answer can still be the wrong answer

Typed interfaces remove parse and schema failures by construction; they do not make the chosen option right. The most direct test held question, state, rubric wording and the set of option names fixed and changed only which name was bound to which rubric [138]. For hosted Jev, rebinding the rubrics behind *no/yes* flipped 32.50% of answers against about 2% under neutral names, 24× the model’s test–retest floor, and lowered AUROC from .8146 to .5806 while the type-error rate stayed at 0%. Swapping the meaning of yes and no flipped about half of Jev’s answers in a decision-gate benchmark [37]. In open models the short label wins over its definition: deleting every definition left one model’s accuracy unchanged, renaming options to A and B gained 0.151, and a label contradicting its definition dropped one model from 0.85 to 0.23 [105]. Supplying a dataset’s definition of hate speech changed up to 28% of predictions without reliably improving them [190].

Two further patterns recur. **Missing answers go unnoticed.** With an explicit “None” option, Jev chose the right value on 99% of arithmetic menus that contained it but correctly rejected only 7% of menus that did

not, although yes/no checks of the same candidates rejected every wrong one in 99% of those cases [181]; when a reference label was removed from the options, Jev noticed 24.8% of the time while an open encoder over-rejected valid menus, and with retrieved menus Jev missed a quarter of out-of-scope queries [191]; Jev gave a hard answer to 63% of items whose deciding evidence had been removed, where a looped open model gave 17.5% [127]; an explicit ABSTAIN option in an open readout caught 28 of 30 invalid menus but answered only 17 of 30 valid ones correctly [111]; on unanswerable medical questions it chose “I don’t know” 10.5% of the time, a frontier model 8.6% [163]. **Middle options absorb errors.** On ANLI, Jev assigned 51% of its errors to Neutral and chose Neutral 37–40% of the time wherever it was placed, and across 36 ordinal datasets decisions used only two-thirds to three-quarters of the gold scale [179].

Order needs care. One community audit saw no argmax flips in 400 order trials on hosted Jev [199], and a 37-dataset benchmark found aggregate accuracy unchanged under rotation [178]; yet four cyclic rotations changed 37.4% of answers item by item on a cyber-knowledge benchmark, 28 points beyond run-to-run variation [15], and the vendor now lists option order among Jev’s weak spots [12]. Aggregate accuracy can hide item-level instability. Consistency is not correctness either: on contract inference, five of Jev’s 30 fixed anchors were answered wrongly in all twelve responses [142]. And an answer can follow the wrong cue: in a 4B model reading trial reports, exchanging intervention and comparator reversed only two-thirds of its direction answers [129].

7.3 F2 · A native probability makes calibration measurable, not guaranteed

Every call returns a distribution, so reliability can be audited on each run, and broad evaluations find it often good: pooled over 22 Choice datasets and 279,925 answers, ECE was 0.028 [178], and on familiar closed-choice tasks Jev was calibrated [184]. Measured reliability still varies by task, construct and language. Across 15 social-science tasks the median ECE was 0.157, better than the verbalised confidence of 16 of 19 LLMs but not of three frontier models, with a task ECE of 0.538 on empathy [23]; in political-science replications Jev was better calibrated than one commercial model’s token probabilities but not consistently better than an open model’s [194]. Probabilities rank better than they threshold: binary probabilities sat badly against a fixed 0.5, and tuned thresholds raised a legal benchmark’s micro-F1 from 0.50 to 0.75 [178]; zero-shot failure detection ranked well (median AUROC 0.886) while its ECE exceeded a null baseline [146]. Thresholds do not travel: one chosen for 5% in-scope risk still admitted 31% of out-of-scope requests, and temperatures fitted on few options raised error with many [37]; a rejection threshold fitted on one task rejected 69% of covered inputs on another [191].

Confidence is also not a signal of missing knowledge. With no answer-relevant information Jev still put up to 0.80 on a salient option, and on news past its knowledge boundary confidence exceeded accuracy by 0.21–0.33 even after recalibration on earlier months; targeted yes/no questions about whether the evidence suffices separated cases better than confidence [184]. On items where 100 annotators split, Jev’s mean Choice confidence was 0.81 against 0.47 agreement among the annotators [200]. Recalibration on held-out target labels helps substantially [17, 141]: one pooled temperature brought an open contrastive model’s calibration error from up to 0.40 to at most 0.062 [45]. Decision-native training improved calibration over a generative twin in medical imaging [123], a 4B trial-reading model met a calibration bar registered before its test [129], and outside language, single-pass stability forecasts held up in a registered prospective test [103]. The background literature explains why per-workload checks are needed [62, 201].

7.4 F3 · A bounded first pass with confidence-gated escalation

Across judging, annotation, moderation, security, infrastructure and agents, the typed model is usually the cheapest and fastest component and rarely the most accurate. The strongest designs now freeze thresholds in advance and test on new workloads. As a judge, Jev came within three points of GPT-6 wherever the verdict could be read off the text; a cascade with a threshold frozen before evaluation beat GPT-6 by 0.9 points on 1,610 held-out pairs at 41% of its fee, and in a pre-specified live test on two new workloads it matched GPT-6’s accuracy exactly while escalating 74% of pairs [14]. In water-network alarm triage, four sealed, pre-registered rounds showed that accepting only benign Jev verdicts confirmed by a rule tree spared an LLM reviewer 35–38% of windows without loss of macro-F1, and the unchanged gate stayed non-inferior to its reviewer on two further networks [189]. A blind-coder audit of population-scale crash-narrative reading found sensitivity 0.999 [36]; an intrusion pilot found F1 0.859 at one example per category [185]; in hate-speech moderation the best hosted decision model came within 1.6 macro-F1 of the best commercial LLM at about 97% lower cost [190]. In moderation, Jev’s confidence kept 72% of one benchmark’s decisions at 94% accuracy [195], and for open decision models, escalating the least confident items to a larger sibling beat every scheme that re-asks the same model at matched cost [133]. Earlier results point the same way in agents and annotation [23, 38].

Every gain is conditional. Trained classifiers win when labels exist: tree ensembles trained on 8,000 flows reached 70% on packet-level traffic classes where Jev with 40 examples reached 28% [180], and a classifier first stage escalating to Jev matched Jev’s accuracy at 0.43 of its cost [37]. A gate cannot rescue a weak first stage: calibrated confidence on a near-chance judge escalated 92–100% of items [45]. Judges can share the typed model’s confident errors: LLM judges repeated 96% of Jev’s most confident errors on rubric criteria, so cascades gained at most 2.7 points even with oracle thresholds [147]. Audited savings shrink: one study found that counting the pre-screen cost turned a reported 23.9% saving into 4.3% [186], and at batch prices Jev had no cost advantage over a commercial LLM because every option is billed as input [194]. Asking targeted questions can beat reading confidence [184]. The lesson for RQ4 is to include cheap trained classifiers and generative cascades as baselines, to fix thresholds on separate data, and to count every cost the gate adds.

7.5 F4 · Speed and cost claims decompose into different mechanisms and scopes

Speed figures measure different things (Table 4). Single-request latencies range from 30.9 ms for a laptop-GPU open model [113] through 0.15 s for hosted judging [14] to 0.75 s for packet classification [180]. Amortized figures divide a batch over questions that share one state [100, 120]. Deadline attainment under load is another quantity: Jev met a 1 s RAN budget on 99.8% of calls where two hosted LLMs met it on 17.9% and 0%, but under load a small self-hosted JSON model also kept up [35], and hosted Jev never met 100 ms near-real-time classes [159]. Caching repeated descriptions gave all interpreters nearly the same latency, so the gain lay in fresh decisions [34]. Inside an agent, hosted Jev took 1.4–2.1 times as long as local scoring, including API communication [182], and a faster step made a slower agent: Jev action selection cut median episode time from 267 s to 161 s but lowered success from 77.6% to 66.1% and lengthened the slowest runs [124]. Billing every option as input can erase the price advantage against batch-priced LLMs [194], and the bill grows with the workload: agent-built task-specific artifacts reached 94–97% of Jev’s quality at 2–25% of its projected full-workload cost [198]. A multi-agent framework that hands coordination decisions to an open decision model used 78–94% fewer GPT tokens than five other multi-agent systems [197]. For RQ3, no single speed or cost number describes a typed decision model; each figure is valid only within its scope.

7.6 F5 · Interface compatibility is not mechanism equivalence

§6 established that the interface is served by many mechanisms. The evidence adds four points. First, frozen readouts are strong baselines: a frozen 4B model matched community Jev-style models on the same backbone [106], and in an earlier same-backbone ablation a typed head gave no consistent gain over reading the LM head [120]. Second, rankings between hosted and open models change with the task: hosted Jev beat an open encoder on 9 of 11 agent-harness decision points [186] and on safety judging [156], while specialised open models beat Jev on their own tasks [118, 119] and a language-specific encoder matched it on a general Chinese subset with better calibration [101]. Rankings shift again in moderation, where retrieved precedents helped Jev and barely moved Laya [195], and on a broad typed-decision suite hosted Jev outsourced every model a looping study trained [127]. Third, reproducibility differs: repeating 1,080 decisions a day later changed 0.5% of hosted answers, while an open checkpoint reproduced 1,180 of 1,180 bit-for-bit on a second machine [186]. Fourth, projects state their own limits: SemIf that it does not reproduce Jev’s model or training, LocalJev that it is wire-compatible but not mathematically equivalent to a logit read, and Kev that its architecture follows community speculation [19, 26, 132].

7.7 F6 · System-level gains need module-level attribution

Systems improve for many reasons at once. A semantic database finished joins with up to 540,000 candidate pairs where baselines timed out, but pruning removed 87.4% of pairs before any decision was made [188]. A search agent cut active search time 3.7–4.7-fold and raised accuracy from 45% to as much as 54% by letting a small model take confident short decisions [118]. Memory systems and navigation agents report gains measured against different baselines [134, 169, 173]. In a multi-agent system, the System One controller beat random choices from the same legal menus by only 1.8 points at similar token use, so its savings came from the framework around it rather than from the quality of its decisions [197]; exposing only the current stage’s candidates raised correct robot decisions from 90 to 113 of 120 [111]; and in semantic databases, plan choice as well as the decision model determines output quality [196]. The clearest attribution comes from letting code own what the model cannot do: Jev solved 31% of ALFWorld games alone and 87% with a code-supplied lookahead, and stating an opponent’s move raised game accuracy from 33% to 98% [187]. A faster step can still cost the episode: replacing an agent’s action selection with Jev lowered success as step errors compounded [124]. Attribution also needs references that measure what the workflow actually

Table 4: Speed and cost figures grouped by measurement scope. Figures in different groups are not comparable.

Scope	Reported figure	Ref.
Single request	Full semantic correctness; median 0.335 s, p95 0.442 s	[136]
Single request	64.5 ms/decision; 4.9× faster than same backbone generating	[33]
Single request	30.9 ms per decision on one laptop GPU; 32 decisions/s	[113]
Single request	Median decision latency −22.7% to −64.5% vs the fastest LLM	[34]
Single request	Self-reported 81.65% on public JevBench items at 7.08 ms	[137]
Single request	Meets the 1 s RIC budget on 99.8% of calls (LLMs: 17.9%, 0%)	[35]
Single request	Single-token LLM classifier: 85.5% at 0.53 s, flat across label spaces	[174]
Single request	Open encoder–decoder: 78.05% on DecisionBench, 32 questions in 221 ms	[100]
Single request	Strong reranking quality; latency between local models and pointwise LLMs	[183]
Single request	0.75 s vs 6.0 s per request; accuracy difference not established	[180]
Single request	Intrusion pilot: F1 0.859, 4.8× faster than an LLM	[185]
Single request	Wireless control: 3.5–8.5× faster than LLMs; quality depends on the task	[192]
Single request	Order invariance by construction; sequential heads still flip 2–3%	[43]
Single request	Reranking: ties Cohere Rerank 4 Pro at half the latency	
Single request	Local 4B readout: 76% on public JevBench at 48 ms	
Amortized per question	8.9× vs serial, 3.4× vs no-reuse batch; 5.7 ms/question amortized	[120]
End to end	0.36% of GPT-6’s fee at 0.15 s median	[14]
End to end	95% success with 1.12 strong calls/task vs 88% and 4.10 (−72.7%)	[38]
End to end	Behind best LLM on 14/15 tasks (−11.6 F1) at 44× lower cost	[23]
End to end	Low-confidence routing matches the LLM at ¼–½ of its cost	[23]
End to end	499,500 narratives screened for \$25.23; 195,857 fully coded	[17]
End to end	LoCoMo judge 0.777 (+11.0%); build 158 s; 0.93 s/query	[134]
End to end	With caching, latencies converge: the gain is in fresh decisions	[34]
End to end	\$0.000228 and 1.24 s per contract — lowest of ten models	[142]
End to end	\$0.0227 per 1,000 predictions; median 213 ms	[140]
End to end	Jev 0.422 s median; \$0.15 of \$3.71 per game	[139]
End to end	Live admission: 0.91–0.95 exact and on time where LLMs fall below 0.1	[34]
End to end	Under load, slow interpreters saturate — a small local JSON model keeps up too	[35]
End to end	One run each: 5 min faster, different severities — not a controlled test	[145]
End to end	AndroidWorld 79% (vs 84% step-wise) at 73% lower API cost	[148]
End to end	Hosted Jev never meets 100 ms near-RT deadlines (warm p50 281 ms)	[159]
End to end	Agent with spawned finite actions: Maze 0.96 vs 0.52	[182]
End to end	Hosted Jev inside the agent: better on 3 of 8 tasks, 1.4–2.1× slower	[182]
End to end	Let code simulate: ALFWorld solved 31% → 87%	[187]
End to end	Semantic SQL: pruning + typed decisions finish 540K-pair joins	[188]
End to end	Pre-registered gate: a third of SCADA reviews offloaded, accuracy kept	[189]
End to end	Self-audit: a 23.9% saving was really 4.3%	[186]
End to end	Faster steps, slower agent: success 77.6% → 66.1%	[124]
End to end	Search agent: 3.7–4.7× faster, accuracy 45% → 54%	[118]
End to end	Browser tasks: specialised open model 38.5% vs Jev 16.7%; single steps alike	
End to end	Routing subagents with Jev: cheaper than one baseline, dearer than another	
End to end	Laya coordinates agents with 78–94% fewer GPT tokens; random choices trail by 1.8 points	[197]
Simulated	Frozen cascade: +0.9 points over GPT-6 at 41% of its fee	[14]
Simulated	Radio conditions outweigh the interpreter; no LLM gain over Jev	[35]
Simulated	Emulated router saves 13–21% of coding-agent spend	[144]
Simulated	Calibrated confidence prices query plans; overconfidence hides output errors	[196]
Author estimate	\$0.000217 electricity per suite pass vs \$10.636 for a hosted model	[113]
Author estimate	Under \$0.03 per 100 report pairs (judgment calls only)	[141]
Author estimate	Agent-built artifacts: 94–97% of Jev’s quality at 2–25% of its projected cost	[198]
Vendor claim	No type errors by construction; plotted 0% is analytical	[3]
Vendor claim	\$0.042 / M input tokens; 70–500 ms end to end (vendor)	[3]
Vendor claim	193.6× faster, 444.6× cheaper on vendor workflow evals	[3]
Vendor claim	Vendor dashboard: 67.8% agreement at \$0.0004 and 0.4 s per case	[20]

uses: administrative codes understated fidelity to the narratives by a median 0.26 in kappa [17], and wrong intermediate selections changed downstream counts while the final label stayed correct [136].

7.8 F7 · The evidence is young, clustered and unreplicated

The core preprints appeared within three weeks of the launch, mostly as a single version, and several were substantially revised soon after: one edge-orchestration study was re-scoped into a different paper, and a judging study's cascade figures changed between versions [14, 35]. Several author groups contribute more than one study (§3), and in one case the author of a benchmark also ranks their own model on it, with results the author flags as not blind [42, 43]. Hosted answers drift: a re-run changed the Score value on 38.5% of Score questions in one audit [153], and 0.5% of decisions a day later in another [186]. Grading a model with its own labels inflates its gain: a reranking gain of +0.053 NDCG under Jev-made labels became -0.028 under an unrelated model's [202]. One study's audit of its own pipeline found three analysis errors that had distorted its headline deployment claims [186]. Practice is improving: several newer studies froze a preregistration or registered criteria before testing [45, 103, 129]. Still, none of the core results has been independently reproduced, and some claimed releases could not be located (§9).

7.9 F8 · A calibrated probability is not a coherent belief

Average calibration says nothing about whether probabilities obey basic rules across related questions. Jev's probabilities that "the label is X" and "the label is not X" missed summing to one by 0.064 on average, about five times its repeat noise, and three single-label probabilities summed to 1.14 [155]. Asking the same probability question through two interfaces changed the act-or-defer decision on 32.8% of pairs [160]. Staged decisions through categories disagreed with direct ones, costing 22.9 points on one intent benchmark [162]. On items with analytically known distributions, Jev picked the most likely option every time but put 89% of the mass on it where the true chance was 36% [124]. Against known probabilities, yes/no and Score answers were close but Choice answers were not [172]. The open readouts tested were less coherent still [155, 162]. Ranking and probability can also part ways: after a change of candidate menu, first-pass probabilities overstated an open decision model's accuracy by 21 points although the top-ranked candidate rarely changed [133]. Probabilities that rank well can still be used for selection; treating them as beliefs to combine requires coherence checks first.

7.10 F9 · Typed decisions read what the input states; they do not reliably compute what it implies

Studies of very different tasks meet the same boundary. Jev answered 99% of Cognitive Reflection Test questions yet played one-shot games as if a rational opponent moved at random [187]. It landed within one of exact numerical answers on 97% of items but exactly on them on 44% [124], read every edge of a graph but counted vertex degree right on 68% [41], followed chains of statements about who lies perfectly at one step but at chance from six [127], classified encrypted traffic from packet sizes and timings at 28% with 40 examples [180], recognised activities from sensor values far below supervised models [176], and rejected arithmetic menus lacking the right value only 7% of the time [181]. Medical questions answerable from an abstract were at parity with a frontier model while diagnostic cases trailed by over 20 points [163]. Range refinement through a sequence of choices recovered some numeric precision [143]. The vendor's guidance agrees: count, convert and compare in code, and keep the model for the judgment [12]. One qualification: calculation-heavy MMLU subjects were easier for Jev than its other subjects, the reverse of two open models, and the authors cannot rule out memorised question-answer pairs [178].

7.11 F10 · A typed output does not make a decision safe from its inputs

Because the answer depends on the state, text placed in the state can move it. How far depends on the attack and the task. In one study, injected content raised the attacker's target probability but selected it in only 1.8% of cases, 3.5% with adaptive search [39]. In another, context-ignoring injection reached 96.5% attack success on support routing, and injections that declared the state outdated and added false information reached 100% on subscription cancellation and at least 96.8% on issue severity [193]. Optimised natural context redirected 61% of correct decisions [150], and an appended observer opinion flipped 12% of decisions and pushed 38% of confident answers below a 0.8 gate [153]. Open checkpoints add a supply chain: poisoning 5% of an open model's training records implanted backdoors that redirected most targeted inputs with clean accuracy intact [193]. Confidence gates themselves are attack surfaces: perturbations that lower confidence

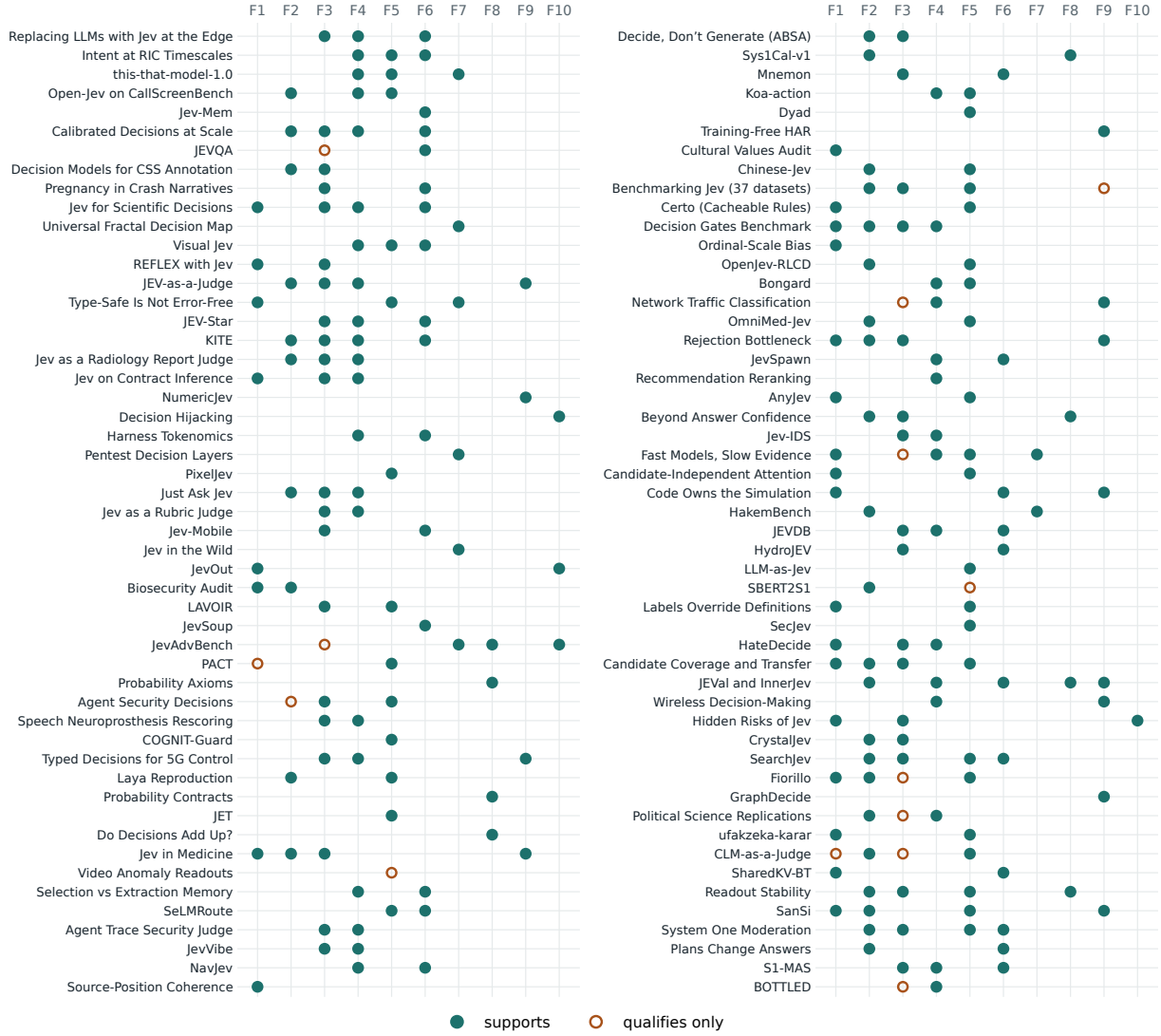


Figure 2: Which study bears on which finding (F1–F10, §7), derived from the evidence records (filled: supports; open: qualifies only).

can drain an escalation tier [203, 204]. The same capability serves defence — as a detector, Jev reached ROC-AUC 0.95–1.0 on eight injection, jailbreak, harmful-content and AI-text benchmarks [193] — but a detector’s own decisions remain manipulable. Constrained outputs limit what an attacker can make the model say, not what it decides.

8 Reliability and failure modes

Table 5 collects the failure modes reported so far, the stage where each arises, a mitigation and its sources. Most mitigations are procedural rather than architectural: neutral identifiers with meaning in the rubric; explicit exits verified by separate yes/no questions; per-item flip rates under rotation; complementary questions to check coherence; thresholds fitted on held-out data from the deployment distribution; arithmetic, counting, dates and simulation in code; trusted and untrusted state kept apart; pinned checkpoints; and episode-level evaluation of agents. Some have been tested on hosted Jev within single studies — held-out thresholds, yes/no verification, rule-confirmed acceptance — and others only on open models; none has been replicated.

Table 5: Failure modes reported so far, with mitigations (none replicated; several tested only on open models).

Failure mode	Stage	Mitigation	Sources
Option names outweigh the rubric	Decision tract	con- Neutral identifiers with meaning carried in the rubric; name-invariance checks next to type-error rates.	[37, 138]
Labels outweigh definitions	Decision tract	con- Test with neutral labels and with labels that contradict their definitions; render options so the definition, not the label, carries the rule.	[105, 190]
Order flips hide in aggregates	Decision tract	con- Report per-item flip rates under rotation next to aggregate accuracy; average over rotations or isolate candidates.	[12, 15, 107, 111, 125, 199]
The primitive changes the answer	Decision tract	con- Fix the primitive in a protocol; test both when a result depends on it.	[160, 172, 199, 205]
Missing answers go unnoticed	Decision tract	con- Keep an explicit exit, verify candidates with separate yes/no questions, and fit a rejection threshold on held-out answer-absent cases.	[127, 163, 181, 191, 199]
Middle options absorb errors	Decision tract	con- Check how much of the label scale predictions use; compare ordinal and nominal framings of the same items.	[179]
Injected state redirects decisions	Decision tract	con- Separate trusted from untrusted state, verify facts in code, and red-team with false-state injections, not only instructions.	[39, 150, 153, 193]
Vendor-listed jagged edges	Decision tract	con- Keep arithmetic and date logic in code; filter state; split indirect questions; re-check answers under reordered options.	[12]
Consistent is not correct	Decision tract	con- Score every repeat against gold labels and report stable errors separately from changing ones.	[142]
Same request, different answer	Inference readout	& Measure a test–retest floor and report effects relative to it.	[138, 153, 186, 199]
Cacheable scoring loses the rules	Inference readout	& Keep the state and rules in every candidate’s view; re-measure rule sensitivity after any caching change.	[125, 126]
Poisoned open checkpoints	Inference readout	& Pin checkpoints by hash and source, and prefer releases whose training data and recipe are documented.	[193]
Confident on the wrong construct	Probability calibration	& Audit reliability per task and construct, not with one global ECE.	[23, 206]
Probabilities that do not add up	Probability calibration	& Ask complementary questions and check sums; use rankings rather than raw magnitudes where coherence was not tested.	[124, 155, 160]
Confident beyond what it knows	Probability calibration	& Ask targeted yes/no questions about whether the evidence suffices; do not read low confidence as missing knowledge.	[184]
Unknowable rules and distribution shift	Probability calibration	& Hold out domains and rule changes; recalibrate on target-domain labels.	[18]
Language shift costs accuracy	Probability calibration	& Evaluate in the deployment language; pin the model version.	[4, 177, 207]
Two-decimal probabilities	Probability calibration	& Treat rare-event probabilities with care; smooth before log-loss.	[17, 18]
Thresholds that do not travel	Selective control	con- Fit thresholds on held-out data from the deployment distribution, per option count, and re-check after any version change.	[37, 178, 186, 191]
Asked to compute, it estimates	Downstream use	Compute in code and ask the model to judge the result; supply simulations or lookahead explicitly.	[41, 124, 127, 180, 187]
Fast steps, slower agent	Downstream use	Evaluate at the episode level, including tail latency and success, not per step.	[124, 182]
Validity effects masquerade as quality	Downstream use	Count invalid outputs as errors and report parse rates separately.	[14]
Right label, wrong quantity	Downstream use	Score intermediate relations and derived quantities, not only final labels.	[136]

9 Openness and reproducibility

Openness is not a single property (Figure 3; Appendix C). Among the 102 core and peripheral studies, code was located in full for 55, in part or as a project page only for 3, and claimed but not located for 3. Weights were located for 11 studies and raw predictions for 18. Data restrictions are sometimes legitimate: police narratives cannot be redistributed under their data agreements [17, 36]. Several studies release everything needed to recompute their numbers, including every raw response [23, 178, 186, 189].

The open ecosystem is larger than the literature (Table 6). We audited 45 open models and readouts, 33 benchmarks and evaluations, 43 applications and 9 tools and catalogues. Among the open models, 22 release weights of their own and 16 have none to release, mostly because they read existing checkpoints; training code was located for 21 and evaluation material for 24. Benchmarks range from public leaderboards to single-author harnesses; some are maintained by teams that also publish a ranked model [42, 208]. Applications mostly report demonstrations or their builders’ own measurements, and some publish candid negative re-

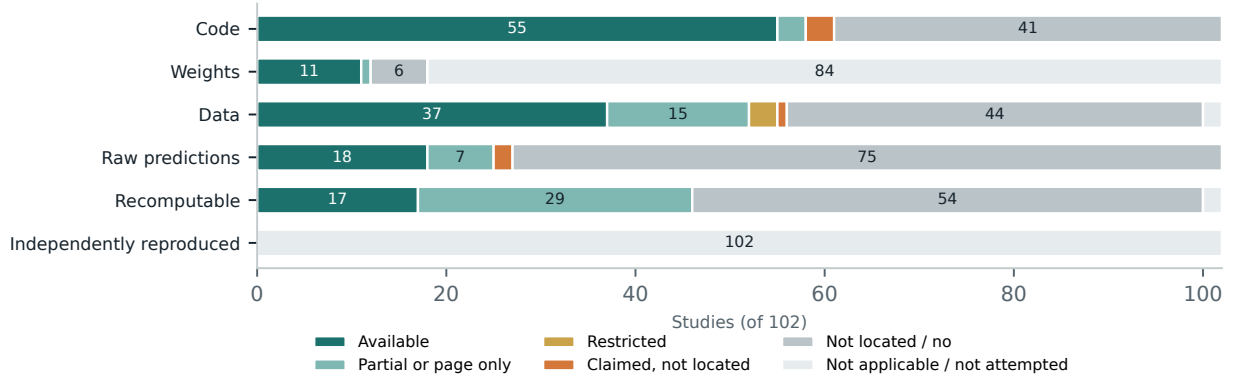


Figure 3: Openness of the core and peripheral studies: number of studies per status in each of six independent fields.

sults, such as a skill router its author removed after a week because the log could not separate its effect from the agent’s own choices [209]. Licence fields reported by GitHub as NOASSERTION mean “not identified”, not “no licence”. We record “not located” rather than “absent” throughout.

Table 6: Open models and readouts, read from their READMEs at pinned commits (nothing executed). “n/a”: no weights of their own, e.g. readouts of existing checkpoints; “not loc.”: not located, which is not the same as absent.

Resource	Family	Base model	Weights	Training	Eval.	Data	Ref.
alperiox/audio-jevlike	Encoder decision heads	Whisper / WavLM audio encoders	not loc.	yes	yes	not loc.	[210]
Contrastive-LM/CLM	Encoder decision heads	Qwen3-8B (frozen encoder)	yes	partial	partial	not loc.	[104]
Heman10x-NGU/Verdict-open-jev	Encoder decision heads	ModernBERT (151M, GLiClass lineage)	yes	yes	yes	not loc.	[211]
hiroki-abe-58/sokudan	Encoder decision heads	sbintuitions/modernbert-ja-310m	yes	yes	yes	not loc.	[102]
mizorewww/laya-mlx	Encoder decision heads	Laya checkpoints (ModernBERT / mmBERT)	yes	not loc.	yes	not loc.	[212]
NandhaKishorM/laya	Encoder decision heads	ModernBERT-large; mmBERT-base	yes	not loc.	partial	not loc.	[99]
olanotolu/jevbetter	Encoder decision heads	From scratch (hashed n-gram encoder)	not loc.	yes	yes	not loc.	[213]
SamratDuttaOfficial/WaterSheep	Encoder decision heads	answerdotai/ModernBERT-base	yes	yes	yes	not loc.	[214]
daseinlabs/open-jev	Frozen decoder option readout	Gemma 3 4B	n/a	yes	partial	not loc.	[27]
ekzhang/openjev-sglang	Frozen decoder option readout	Qwen3.6-35B-A3B	n/a	not loc.	partial	not loc.	[215]
featherless-ai/simple-jev	Frozen decoder option readout	Open instruction-tuned models (e.g. Gemma 4 26B-A4B)	n/a	not loc.	yes	not loc.	[109]
nokia-applied-research/AnyJev	Frozen decoder option readout	any causal LM (Qwen3-8B in README)	n/a	not loc.	partial	not loc.	[216]
rorshopping/jev-on-a-laptop	Frozen decoder option readout	Stock 1.5B–8B open models	n/a	not loc.	yes	not loc.	[217]
TheoLeeCJ/SemIf	Frozen decoder option readout	frozen Qwen3.5-4B and others	n/a	not loc.	partial	not loc.	[26]
Yinsongxu/LLM2Jev	Frozen decoder option readout	Qwen3.5 (e.g. 4B) and other local LLMs	n/a	not loc.	yes	not loc.	[108]
zhengxuyu/litjev	Frozen decoder option readout	Qwen model family	n/a	not loc.	not loc.	not loc.	[110]
zhihz/openjev	Frozen decoder option readout	Qwen3-4B-Instruct-2507 (frozen)	n/a	not loc.	yes	not loc.	[28]
bespokelabsai/nimble	Fine-tuned decoder decision models	Qwen3.5-9B (LoRA)	yes	yes	not loc.	yes	[115]
FLock-io/this-that-model	Fine-tuned decoder decision models	decider-2b (Qwen3.5-2B lineage)	yes	not loc.	yes	partial	[218]

Resource	Family	Base model	Weights	Training	Eval.	Data	Ref.
getainode/jebadiah	Fine-tuned decoder decision models	Qwen3.8-27B, Qwen3.5-9B, Qwen3.5-4B (LoRA)	yes	yes	yes	yes	[219]
gitchw/LCT	Fine-tuned decoder decision models	Qwen2.5-0.5B, Qwen2.5-1.5B, Qwen3-8B	yes	yes	yes	yes	[128]
guanxuyu-sv/Visual-Jev	Fine-tuned decoder decision models	Qwen3-VL-4B	n/a	not loc.	not loc.	not loc.	[220]
iapp-technology/openthai-systemone	Fine-tuned decoder decision models	Qwen3.5-0.8B (continued pre-training on Thai)	yes	yes	not loc.	not loc.	[116]
jaredpalmer/kev	Fine-tuned decoder decision models	Qwen3.5 (0.8B, 4B, 9B)	yes	yes	yes	yes	[19]
lexmount/WebJev	Fine-tuned decoder decision models	Qwen3.5-35B-A3B-Base	yes	yes	yes	yes	[119]
Mapika/decider	Fine-tuned decoder decision models	Qwen3.5-2B/4B/35B-A3B	yes	partial	not loc.	not loc.	[112]
nico-martin/open-jev	Fine-tuned decoder decision models	Kev and DeBERTa-v3 ONNX exports	yes	not loc.	not loc.	not loc.	[32]
OmniJev/OneJev	Fine-tuned decoder decision models	Qwen3.5-0.8B to Qwen3.8-27B (vision tower kept)	yes	yes	yes	yes	[121]
OmniJev/PlayJev	Fine-tuned decoder decision models	Qwen3.5-0.8B-Base	yes	yes	not loc.	yes	[122]
PsiACE/dohnuts	Fine-tuned decoder decision models	0.8B multimodal base (LoRA)	yes	yes	yes	not loc.	[221]
Rizzo-AI-Academy/rizzo-flow	Fine-tuned decoder decision models	Spark-X2.5 4B and 1.7B (LoRA)	yes	yes	yes	not loc.	[222]
shamazharikh/qwen-rlcd	Fine-tuned decoder decision models	Qwen3.5-0.8B	not loc.	partial	not loc.	not loc.	[223]
TianyuCodings/NanoJev	Fine-tuned decoder decision models	0.6B decoder	yes	not loc.	partial	yes	[114]
uspraveen/Jevify	Fine-tuned decoder decision models	Gemma 4 E4B, Qwen3.5-2B/9B, Qwen3-VL-2B	yes	yes	yes	yes	[224]
Zefan-Cai/Open-Jev	Fine-tuned decoder decision models	Open-Jev 2B, 9B and 27B checkpoints	yes	yes	yes	yes	[31]
JoshuaSP/open-jev	Diffusion structured readout	DiffusionGemma 26B-A4B	n/a	not loc.	yes	not loc.	[30]
mmastrac/djev	Diffusion structured readout	DiffusionGemma	n/a	not loc.	not loc.	not loc.	[130]
razorback16/openjev	Diffusion structured readout	DiffusionGemma 26B-A4B	n/a	not loc.	not loc.	not loc.	[29]
githubnext/localjev	Generative & constrained adapters	DiffusionGemma via chat completions	n/a	not loc.	not loc.	not loc.	[132]
typesafe-ai/system-one-adapter-python	Generative & constrained adapters	—	n/a	not loc.	not loc.	not loc.	[131]
feder-cr/jev	—	—	not loc.	not loc.	yes	not loc.	[225]
PAI-CUHK/MEDJEV	—	Qwen3 backbones (e.g. Qwen3-0.6B)	not loc.	yes	partial	not loc.	[226]
PAI-CUHK/SLEEPJEV	—	Polysomnography signal encoder	not loc.	yes	yes	not loc.	[227]
pCwOrM/werr	—	—	n/a	not loc.	not loc.	not loc.	[228]

Resource	Family	Base model	Weights	Training	Eval.	Data	Ref.
vinnylarouge/jevlike	—	—	not loc.	yes	yes	not loc.	[229]

10 A proposed evaluation protocol

No experiment is reported in this survey. The protocol below is proposed so that the open questions of §7 can be answered with comparable numbers.

Tasks. Seven compact tracks: intent and semantic classification; pairwise judging; tool and model routing; narrative variable coding; sequential agent control; open-set decisions with the right answer present or absent; and numerically grounded decisions in which the needed quantity must be computed. Visual inputs form a separate track, and text-only systems receive the same extracted features.

Systems and baselines. A majority or rule baseline; TF-IDF or embedding plus a linear classifier; Set-Fit, ModernBERT and GLiClass [51–53]; frozen-LM option readouts with and without rotation averaging; constrained JSON generation with the same backbone [79]; open decision models, general and specialised; version-pinned hosted Jev; a strong LLM; and two cascades (cheap trained or generative first stage, Jev first stage). Supervised systems report their training data and cost.

Metrics. Accuracy or macro-F1 with invalid outputs counted as errors; NLL, Brier and ECE with stated binning and bootstrap intervals, keeping primitives separate; coherence residuals for complementary questions and for the same question asked through different interfaces; missing-answer detection and false rejection; risk–coverage curves and coverage at fixed risk, with thresholds and temperatures fitted on validation data only; per-item flip rates under option rotation, name–rubric rebinding, label–definition conflict, neutral and random names and language variants, each against a test–retest floor; attack success for instruction-style and false-state injections; single-request p50/p95, amortized per-question time, throughput, deadline attainment under load and end-to-end time as separate columns; API bills per correct decision, including every pre-screen and escalation call.

Splits and records. Group splits and bootstraps by scenario or document. For each item keep the input, schema, option order, full distribution, model ID and version, timestamp, errors, retries and cost. Register the analysis plan, fix thresholds before held-out evaluation, and repeat a slice on a later day.

First experiments. (i) Name–rubric binding $\times$ native calibration $\times$ selective escalation on a fixed test set: does low-confidence escalation catch binding-induced errors, and at what cost? (ii) Missing-answer menus $\times$ yes/no verification $\times$ held-out rejection thresholds. (iii) Coherence of complementary questions $\times$ interface $\times$ recalibration. Each asks whether structured, probabilistic errors are in fact easier to intercept ($F1 \times F2 \times F3 \times F8$).

11 Open problems

1. **Semantic invariance by design.** Train or render readouts so that decisions follow definitions rather than labels or option-name polarity, and report name invariance next to type-error rates.
2. **Exits that work.** Make “none of the above” a real decision, not an option the model ignores, and evaluate it on answer-absent menus.
3. **Coherent probabilities.** Enforce or test consistency across complementary questions, granularities and interfaces before probabilities are combined.
4. **Computation boundaries.** Detect when a decision needs counting, arithmetic or simulation, and hand it to code automatically.
5. **Per-construct reliability.** Identify, before deployment, the constructs and languages on which confidence is uninformative, using small labelled pilots and recalibration.
6. **Escalation that transfers.** Find threshold rules that survive distribution shift, option-count changes and a change of fallback model, under explicit risk budgets [71, 72].
7. **Security of the decision layer.** Separate trusted from untrusted state, defend against false-state injection, audit open checkpoints for backdoors, and protect escalation tiers from draining attacks.
8. **Reproducible commercial evaluation.** Pinned versions, raw responses and dated repeats are needed to separate model drift from effects.
9. **Honest efficiency accounting.** Report the scope of every timing and cost figure and attribute gains to readout, batching, caching, pruning and network.

12 Limitations

This survey was carried out by one AI-assisted reviewer, without a second independent screener or a registered protocol. Literature search used the arXiv API (metadata fields), GitHub and curated lists; other databases were not searched exhaustively, and the field changes daily. Background references were checked at abstract depth. Repositories were read, not executed. No result was reproduced, and every number is author-, vendor- or community-reported. Assigning a study to findings, stages and applications involves judgment; the records make every assignment inspectable. The five stages and six families are an analytical organisation, not an architecture. DOI and peer-review status are intentionally unset.

13 Conclusion

Typed decision models make an old requirement concrete: software needs answers it can parse, probabilities it can act on and a principled route to escalate when unsure. Three weeks of evidence on Jev show that the first requirement is met by construction; the second holds on average for many tasks but must be verified per workload, and coherence across questions cannot be assumed; and the third works as a first pass with thresholds fixed in advance rather than as a replacement for stronger models. The same evidence draws a sharp boundary: typed decisions read well and compute poorly, and they are no safer than the state they are given. An open ecosystem already reproduces the interface in many ways and beats the commercial model on specialised tasks, which is a strength for research and a warning against reading any of them as the commercial model. The most useful next steps are version-pinned, same-protocol experiments that measure binding, rejection, coherence and escalation together, with raw predictions released.

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A Search queries

Table 7: arXiv API queries: the 17 accepted snapshot queries (Q) and the expansion queries (X), with hits on 23 September and in the run of 7 Oct 2026. Date-bounded queries were re-run with the upper bound moved to the run date; a query returning implausibly many records was rejected by rule, not screened.

ID	Query	23 Sep	7 Oct
Q01_jev	all:jev	14	86
Q02_typesafe	all:typesafe	5	20
Q03_system_one	all:"System One" AND submittedDate:[202601010000 TO 202609232359]	84	110
Q04_typed_decision	all:"typed decision"	20	44
Q05_calibrated_decisions	all:"calibrated decisions"	46	57
Q06_jevqa	all:jevqa	1	1
Q07_jev_survey	(all:jev OR all:typesafe OR all:"System One") AND (ti:survey OR ti:review)	25	25
Q08_decision_survey	(all:"decision model" OR all:"decision models") AND (ti:survey OR ti:review) AND submittedDate:[202501010000 TO 202609232359]	5	5
Q09_rlcd	all:RLCD AND (cat:cs.AI OR cat:cs.CL OR cat:cs.LG)	2	7
Q10_recent_decision_models	(ti:"decision model" OR abs:"decision model" OR abs:"decision-only" OR abs:"typed decisions" OR abs:"typed questions") AND submittedDate:[202609140000 TO 202609232359]	12	78
Q11_recent_systemone	(all:"system-one" OR all:systemone OR all:"TypeSafe AI") AND submittedDate:[202609140000 TO 202609232359]	6	36
Q12a_openjev	ti:openjev OR abs:openjev	0	1
Q12b_nanojev	ti:nanojev OR abs:nanojev	0	1
Q12c_jevlite	ti:jevlite OR abs:jevlite	1	1
Q12d_jevlike	ti:jevlike OR abs:jevlike	0	0
Q13_rlcd_expanded	all:"Reinforcement Learning for Calibrated Decisions"	0	3
Q14_typed_surveys	(ti:survey OR ti:review OR ti:overview) AND (all:"typed decision" OR all:"decision-only" OR all:"non-generative") AND submittedDate:[202001010000 TO 202609232359]	4	4
X01_machine_native	all:"machine-native" AND submittedDate:[202601010000 TO 202609232359]	3	3
X02_decision_head	(abs:"decision head" OR abs:"decision heads") AND submittedDate:[202606010000 TO 202609232359]	3	8
X03_open_model_names	(all:laya OR all:anyjev OR all:semif OR all:djev OR all:localjev OR all:"this-that-model" OR all:"visual jev") AND submittedDate:[202609010000 TO 202609232359]	rejected	retired
X03a_laya	(ti:laya OR abs:laya) AND submittedDate:[202609010000 TO 202609232359]	0	12

ID	Query	23 Sep	7 Oct
X03b_anyjev	ti:anyjev OR abs:anyjev	0	2
X03c_semif	ti:semif OR abs:semif	1	2
X03d_djev	ti:djev OR abs:djev	0	0
X03e_localjev	ti:localjev OR abs:localjev	0	0
X03f_this_that	ti:"this-that-model" OR abs:"this-that-model"	rejected	rejected
X03g_visual_jev	ti:"visual jev" OR abs:"visual jev"	1	1
X04_system1_model	(all:"System 1 model" OR all:"System-1 model" OR all:"system one model" OR all:"System One models") AND submittedDate:[202609010000 TO 202609232359]	2	26
X05_typed_outputs	(abs:"typed output" OR abs:"typed outputs" OR abs:"typed probabilistic" OR abs:"option probabilities") AND submittedDate:[202606010000 TO 202609232359]	5	10
X06_confidence_gating	(abs:"confidence-gated" OR abs:"confidence gated" OR abs:"selective escalation" OR abs:"escalate when") AND submittedDate:[202606010000 TO 202609232359]	73	96
X07_benchmark_names	(all:jevbench OR all:callscreenbench OR all:"sysone" OR all:"workflow evals")	4	4
X08_decision_models_plural	(ti:"decision models" OR abs:"decision models") AND (cat:cs.CL OR cat:cs.AI OR cat:cs.LG) AND submittedDate:[202609140000 TO 202609232359]	4	49
X09_structured_decisions	(abs:"structured decisions" OR abs:"fast structured decisions" OR abs:"smart if-statements" OR abs:"semantic if") AND submittedDate:[202606010000 TO 202609232359]	rejected	retired
X09a_structured_decisions	abs:"structured decisions" AND submittedDate:[202606010000 TO 202609232359]	42	51
X09b_smart_if	abs:"smart if-statements" OR abs:"smart if statements"	0	0
X09c_semantic_if	abs:"semantic if" OR abs:"semantic ifs"	rejected	rejected

B Evidence record fields

Each record in `data/claims.json` contains: `id`, `subject` (paper, repository or vendor source), `claim_text`, `headline`, `source_url`, `locator`, `evidence_type`, `reported_by`, `independently_reproduced` (false unless a public run log exists), `metric`, `value`, `unit`, `baseline`, `task`, `sample_size` (n and unit, or a reason), `model`, `model_version` (value or reason), `hardware` (value or reason), `test_level`, `measurement_scope`, `findings` (F1–F10 with relation), `limitations` and `note`.

C Openness of the core and peripheral studies

Table 8: Openness of core and peripheral studies: six separate fields (“not located” is not “absent”).

Study	Code	Weights	Data	Predictions	Recomputable	Reproduced
Replacing LLMs with Jev at the Edge	not located	n/a	partial	not located	no	no
Intent at RIC Timescales	not located	n/a	partial	not located	no	no
this-that-model-1.0	available	available	partial	not located	partial	no
Open-Jev on CallScreenBench	not located	not located	not located	not located	no	no
Jev-Mem	available	n/a	available	not located	.	no
Calibrated Decisions at Scale	available	n/a	restricted	partial	partial	no
JEVQA	not located	n/a	partial	not located	no	no
Decision Models for CSS Annotation	available	n/a	available	available	available	no
Pregnancy in Crash Narratives	available	n/a	restricted	not located	no	no
Jev for Scientific Decisions	not located	n/a	partial	not located	no	no
Universal Fractal Decision Map	partial	n/a	partial	not located	.	no
Visual Jev	page only	not located	available	partial	partial	no
REFLEX with Jev	not located	n/a	partial	not located	no	no
JEV-as-a-Judge	claimed	n/a	partial	claimed	no	no
Type-Safe Is Not Error-Free	claimed	not located	claimed	claimed	no	no
JEV-Star	available	n/a	partial	partial	no	no
KITE	available	n/a	partial	not located	partial	no
Jev as a Radiology Report Judge	not located	n/a	restricted	not located	no	no
Jev on Contract Inference	available	n/a	available	available	available	no
NumericJev	available	n/a	not located	not located	partial	no
Decision Hijacking	not located	n/a	not located	not located	no	no
Harness Tokenomics	not located	n/a	not located	not located	no	no
Pentest Decision Layers	available	n/a	not located	not located	no	no
PixelJev	not located	n/a	not located	not located	no	no
Just Ask Jev	available	n/a	available	available	available	no
Jev as a Rubric Judge	not located	n/a	not located	not located	no	no
Jev-Mobile	not located	n/a	not located	not located	no	no
Jev in the Wild	available	n/a	partial	not located	no	no
JevOut	available	n/a	available	partial	partial	no
Biosecurity Audit	available	n/a	available	available	available	no
LAVOIR	available	available	available	not located	partial	no
JevSoup	available	n/a	not located	not located	no	no
JevAdvBench	available	n/a	available	available	available	no
PACT	available	available	available	available	available	no
Probability Axioms	available	n/a	available	available	available	no
Agent Security Decisions	available	n/a	available	available	available	no
Speech Neuroprosthesis Rescoring	available	n/a	available	available	available	no
COGNIT-Guard	available	available	available	not located	partial	no
Typed Decisions for 5G Control	not located	n/a	not located	not located	no	no
Laya Reproduction	partial	available	available	available	available	no
Probability Contracts	not located	n/a	not located	not located	no	no
JET	available	n/a	not located	not located	no	no
Do Decisions Add Up?	available	n/a	available	available	available	no
Jev in Medicine	not located	n/a	available	available	partial	no
Video Anomaly Readouts	not located	n/a	not located	not located	no	no
Selection vs Extraction Memory	available	n/a	available	not located	partial	no
SeLMRoute	available	n/a	not located	not located	no	no
Agent Trace Security Judge	not located	n/a	not located	not located	no	no
JevVibe	not located	n/a	not located	not located	no	no
NavJev	not located	n/a	not located	not located	no	no
Source-Position Coherence	available	n/a	available	not located	partial	no
Decide, Don’t Generate (ABSA)	available	n/a	available	not located	partial	no
Sys1Cal-v1	not located	n/a	not located	not located	no	no
Mnemon	available	n/a	not located	available	partial	no
Koa-action	not located	n/a	not located	not located	no	no
Dyad	not located	n/a	not located	not located	no	no
Training-Free HAR	not located	n/a	not located	not located	no	no
Cultural Values Audit	not located	n/a	not located	not located	no	no
Chinese-Jev	not located	n/a	not located	not located	no	no
Benchmarking Jev (37 datasets)	available	n/a	available	available	available	no
Certo (Cacheable Rules)	not located	n/a	not located	not located	no	no
Decision Gates Benchmark	not located	n/a	not located	not located	no	no
Ordinal-Scale Bias	available	n/a ³⁶	available	not located	partial	no
OpenJev-RLCD	available	n/a	not located	not located	partial	no
Bongard	available	available	not located	not located	partial	no
Network Traffic Classification	not located	n/a	available	not located	no	no